

ADVANCE PROBLEM PROGRAMMING

SLO 2: Lambda and Stream API

1. Use Stream API to Filter and Print Elements from a List

```
import java.util.*;

public class Main {

    public static void main(String[] args) {

        List<Integer> numbers = Arrays.asList(10, 15, 20, 25, 30);

        numbers.stream()

            .filter(n -> n > 20)

            .forEach(System.out::println);

    }

}
```

Output:

25

30

The filter() method selects elements that satisfy the given condition, and forEach() prints them.

2. Write a Lambda Expression to Compare Two Integers

```
import java.util.function.BiFunction;

public class Main {

    public static void main(String[] args) {

        BiFunction<Integer, Integer, Integer> largest =
```

```
(a, b) -> (a > b) ? a : b;
```

```
        System.out.println("Largest = " + largest.apply(10, 20));
    }
}
```

Output:

Largest = 20

The lambda expression `(a, b) -> (a > b) ? a : b` compares two integers and returns the larger value.

3. Chain map, filter, and collect Operations in a Java Program

```
import java.util.*;

import java.util.stream.Collectors;

public class Main {

    public static void main(String[] args) {

        List<Integer> numbers = Arrays.asList(1, 2, 3, 4, 5, 6);

        List<Integer> result = numbers.stream()

            .map(n -> n * 2)

            .filter(n -> n > 5)

            .collect(Collectors.toList());

        System.out.println(result);

    }
}
```

Output:

[6, 8, 10, 12]

Explanation:

- `map()` multiplies each number by 2.
- `filter()` selects numbers greater than 5.
- `collect()` collects the resulting elements into a List.